

FINNOMENA: Helping Thai people achieve their financial goals with Google Cloud

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ABOUT FINNOMENA

With Google Cloud, FINNOMENA has the creativity and ability to design and deploy a robo-advisor platform that enables people to make smart decisions about their financial markets.



About FINNOMENA

FINNOMENA, a financial services knowledge and advisory business based in Bangkok, aims to improve financial well-being among Thailand's 69 million-plus strong population. To achieve its ambitions, FINNOMENA has developed an ecosystem that comprises a robo-advisor platform, a financial content hub and an investment trading platform running on a reliable infrastructure.

Industries: Financial Services &
Insurance

Google Cloud results

- Reduces operational efforts by 80%
- Supports peak loads up to 15x normal traffic volumes
- Fosters creativity and agility through continuous integration and fast deployments

Support growth month users views

Location: Thailand

(<https://cloud.google.com/kubernetes-engine/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/pubsub/>), [Cloud](#)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Functions

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Build (<https://cloud.google.com/build/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Operations

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud Dataproc

(<https://cloud.google.com/dataproc/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Operations

(<https://cloud.google.com/products/operations>)

FINNOMENA (<https://www.finnomena.com>), a financial

services knowledge and advisory business based in

Bangkok, aims to improve financial well-being

among Thailand's 69 million-plus strong population.

"Our vision is to unlock the value and potential of

investments and provide advice on investment

portfolios for Thai people through our 'robo-advisor'

platform, so they can enjoy a more financially secure

retirement," says Attaphong Rattanaveerachanon,

Chief Technology Officer of FINNOMENA.

“We enable people to invest money at the right time in the right assets by providing access to information previously only available to wealthy people,” he adds.

To achieve its ambitions, FINNOMENA has developed an ecosystem that comprises the robo-advisor platform, a financial content hub, and an investment trading platform—and runs it on a reliable infrastructure.

FINNOMENA’s founders initially decided to run its ecosystem in a cloud service. However, as traffic and volumes of content on the platform increased, they realized they needed stability, scalability, and performance at a sustainable cost. “For our business to be successful, we had to continue to attract tens of thousands of new users and hold billions of Thai baht under advisory,” explains Rattanaveerachanon. “We simply could not afford any substantial downtime. At the same time, as a startup, we could not afford to dedicate technology resources to running infrastructure.

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provided a range
of
complementary
services, such
as Stackdriver
logging and
monitoring,
Google Cloud
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—Attaphong
Rattanaveerachanon,
Chief Technology Officer,
FINNOMENA

Evaluation leads to Kubernetes

“Our evaluation quickly led us to Kubernetes container orchestration,” he adds. “The system would

enable us to scale without increasing our operations team and deliver our applications easily regardless of the increasing complexity of our needs."

"Given Google originally designed Kubernetes, and provided a managed service—[Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine/)—that eliminated the need for us to set up and manage our own Kubernetes cluster and provided a range of complementary services, such as [Stackdriver](https://cloud.google.com/stackdriver/), [Logging](https://cloud.google.com/logging/) and [Monitoring](https://cloud.google.com/monitoring/), Google Cloud was the right option for us."

Internal resources used for migration

FINNOMENA elected to move its services one by one from its incumbent cloud provider to [Google Cloud](https://cloud.google.com/) using internal resources. "We adopted a cross-cloud strategy over the three-month migration to minimize the risk of disruption to our customers," says Rattanaveerachanon. "Our infrastructure requirement was relatively uncomplicated and the Google Cloud documentation was clear and concise, so we completed the project smoothly using internal resources."

The FINNOMENA Google Cloud architecture now comprises Google Kubernetes Engine for stateless services that simultaneously serve various business workloads of its ecosystem, such as its robo-advisor service and trading platform; compute resources

delivered through Compute Engine

(<https://cloud.google.com/compute/>) that power some stateful services like the organization's content management system; relational databases on Cloud SQL (<https://cloud.google.com/sql/>); and Cloud Build (<https://cloud.google.com/build/>) that enables continuous integration for the containerized solution across the FINNOMENA development team.

The FINNOMENA analytics data pipeline includes Cloud Pub/Sub (<https://cloud.google.com/pubsub/>) message-oriented middleware and Cloud Dataflow (<https://cloud.google.com/dataflow/>) and Cloud Dataproc (<https://cloud.google.com/dataproc/>) for batch and stream processing. Cloud Storage (<https://cloud.google.com/storage/>) hosts a data lake to retain raw and processed analytical data. The orchestration provided through these services empowers the team to deploy and modify the data pipeline with ease in minutes and delivers the scalability to handle data flows during demand peaks.

Moreover, the team also uses Cloud GPUs (<https://cloud.google.com/gpu/>) to deliver AI and machine learning-powered intelligent investment strategies to customers.

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Chief Technology Officer,
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Democratizing access to information

The scalability, availability, and performance of Google Cloud are helping FINNOMENA realize its ambition to help democratize access to financial information and services in Thailand. The business has increased its number of monthly average users from 5,000 to 20,000 and grown average page views from 20,000 per day to 80,000 per day, while total

assets under advisory had reached 6.7 billion baht (USD\$220 million) by July 2019.

“Google Cloud enables us to be more creative, more agile and complete deployments faster and with few resources,” says Rattanaveerachanon. “This means we can ship products to market more quickly to provide a better service to our customers.”

“Google Cloud
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—Attaphong
Rattanaveerachanon,
Chief Technology Officer,
FINNOMENA

Reduce operational efforts by 80%

The business has reduced its operational efforts by up to 80% through containerization, automated deployment, and logging. Measures that lower cost and effort include using Google Cloud technologies to build a data pipeline, rather than investing the time and effort of internal resources to build one, and moving from a MySQL database running in a virtual machine to Cloud SQL, eliminating the time engineers had to spend optimizing and managing the virtual machine to handle business growth. In addition, FINNOMENA has not experienced any significant downtime due to infrastructure issues since moving to Google Cloud—meaning its platform is available as and when its users need it.

With Google Cloud, FINNOMENA is able to accommodate demand peaks up to 15x greater than standard traffic levels without compromising speed or availability—and without diverting valuable internal resources to manage the increased load.

The value of focusing its internal teams on building new products is shown by FINNOMENA's recent launch of its crowd-sourced robo-advisor investment service. This service allows investors to "follow the lead" of investment experts—including traders, analysts, institutional investors, and asset management companies—in their trading strategies.

FINNOMENA is now well placed to build on its success in the Thailand market. "Google Cloud services enable us to focus on development and solving customer problems and, because they scale

easily, support our growth ambitions,” says Rattanaveerachanon.

**Learn
More**

Reach out to
our team to
see how
Google Cloud
can help your

business.

